

Cooking up holographic BHs

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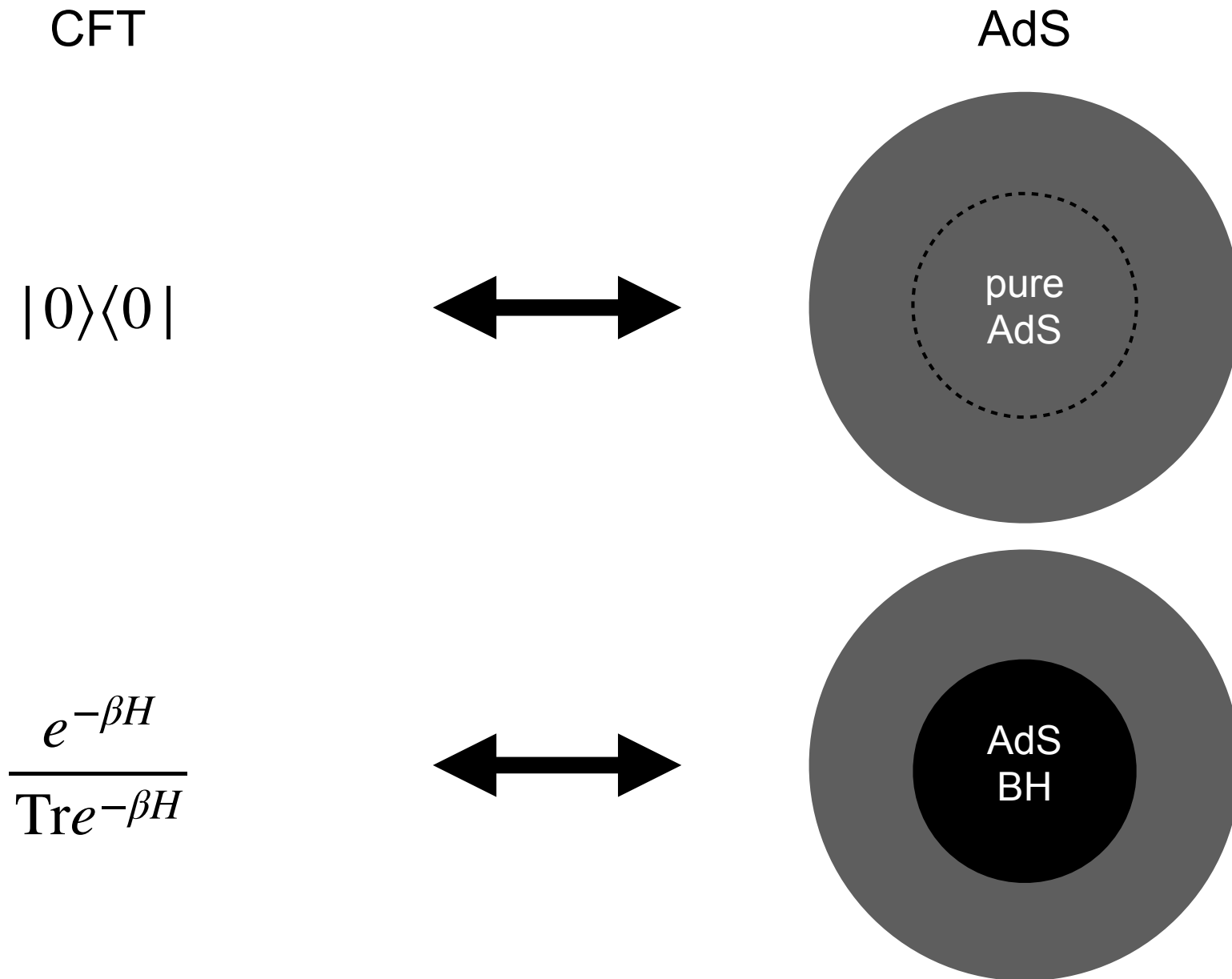
April 27, 2026 @iTHEMS

Based on
Ishii, Takeda [2504.17320] (PRD)
Adachi, Hasegawa, Ishii, Takeda (in preparation)

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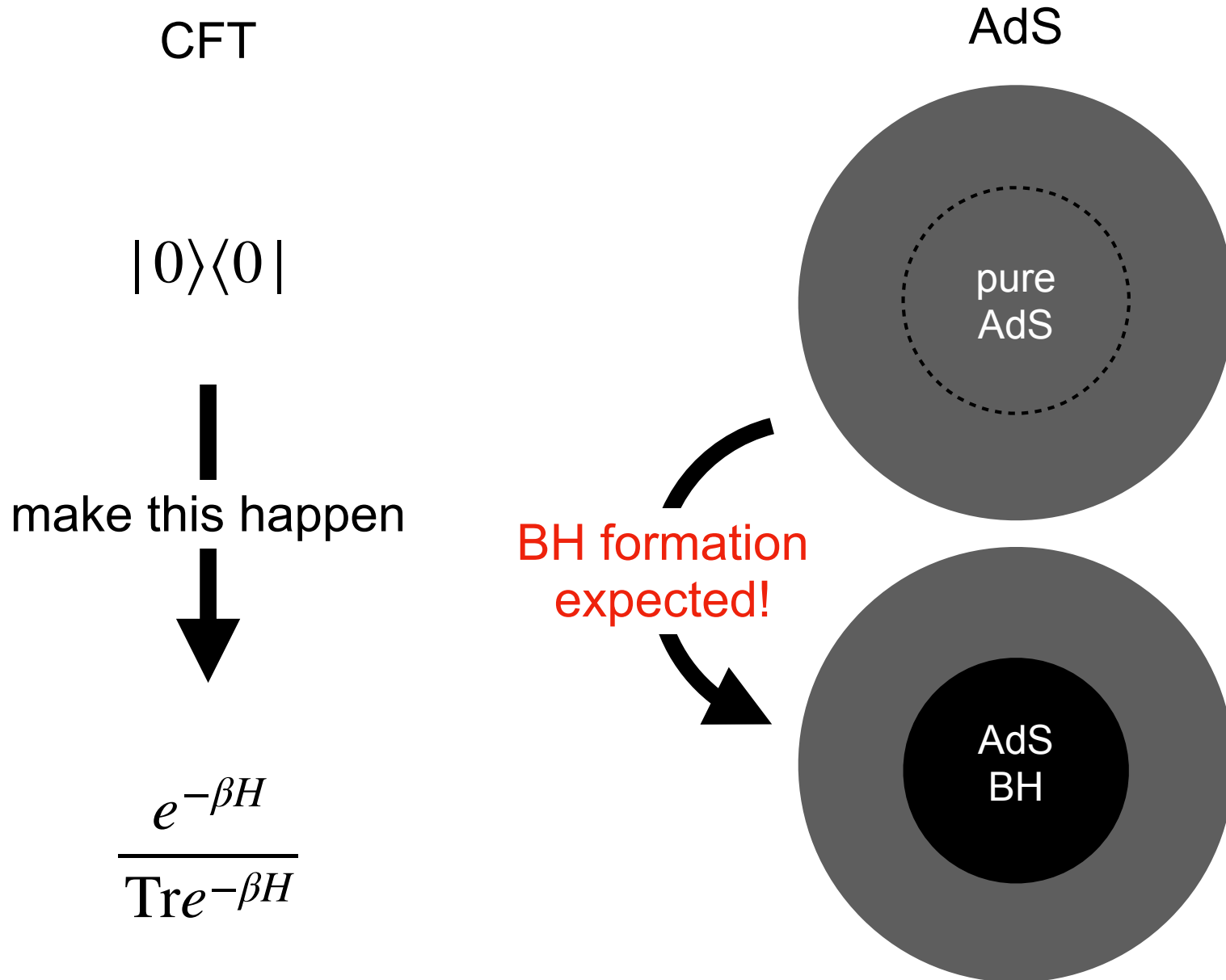
def: to prepare and heat food for a meal

Vacuum and Gibbs in AdS/CFT



What if CFT is open?

Imagine designing bath to thermalize CFT



BH formation should happen always!

Conjecture

For open AdS/CFT, an event horizon forms generally

Key: irreversibility, information loss, entropy increase

Prediction for non-perturbative dynamics!

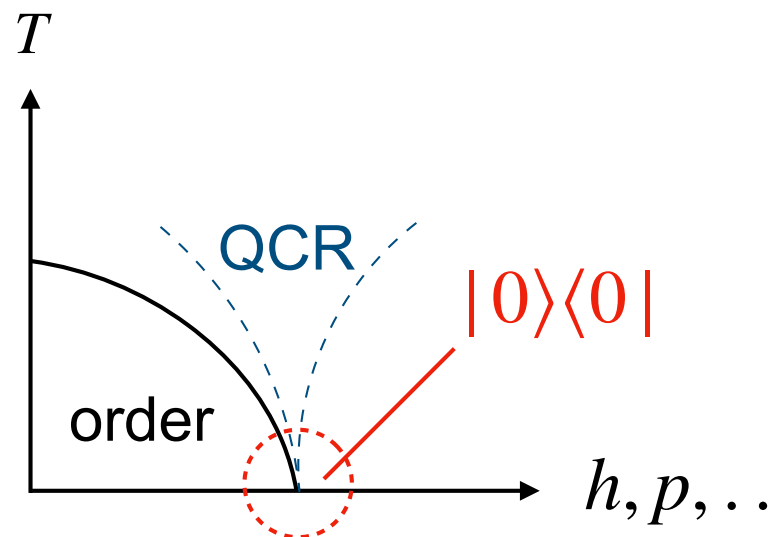
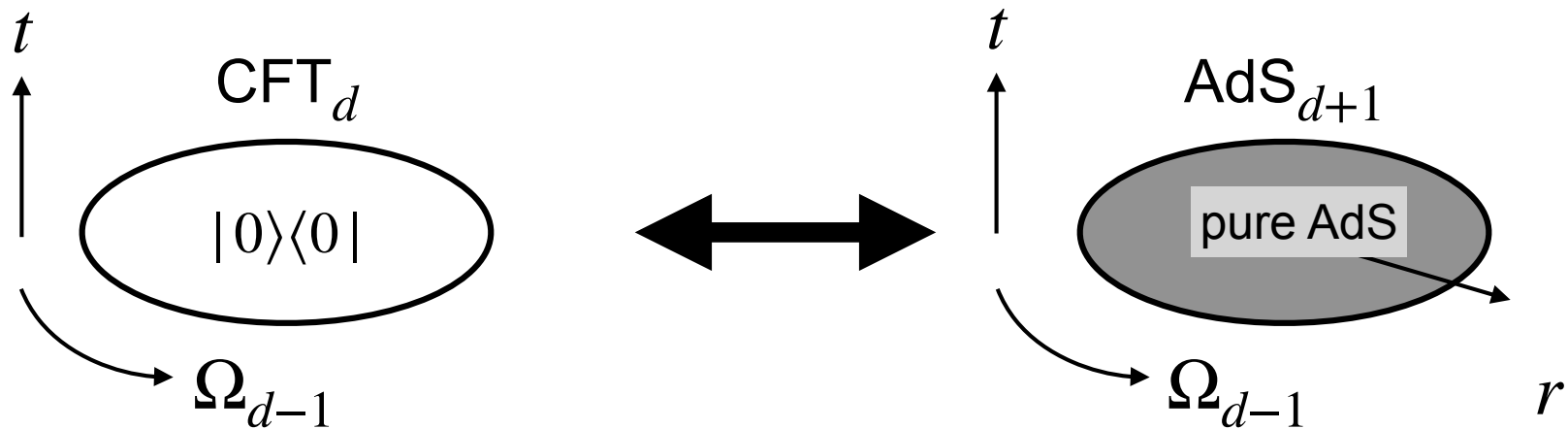
Black hole forms in open AdS/CFT

1. Foundation: closed AdS/CFT
2. Open AdS/CFT dictionary
3. Black hole formation in open JT gravity

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Vacuum vs pure AdS

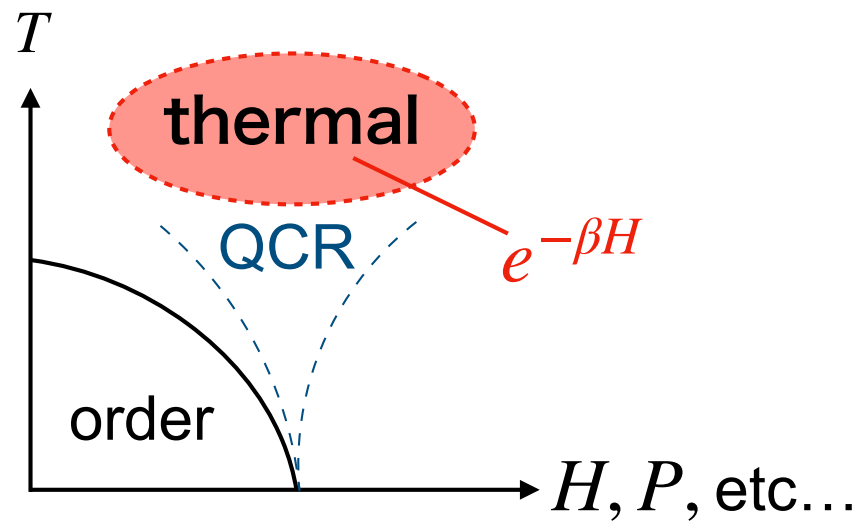
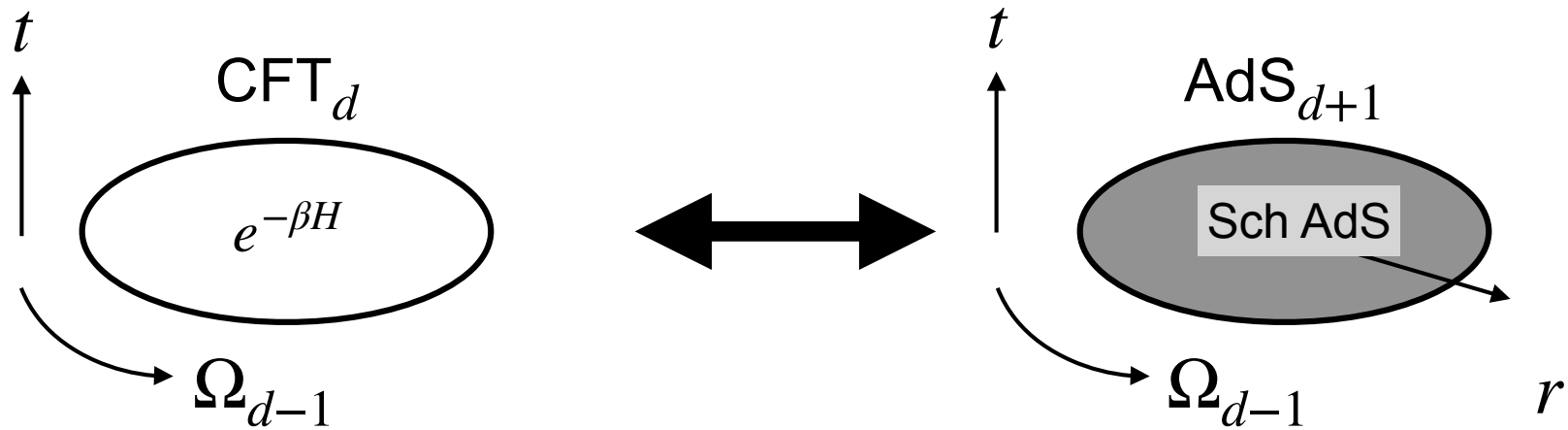


Max sym sol of vacuum Einstein eq

$$R_{\mu\nu} + \frac{R}{2}g_{\mu\nu} + \underbrace{\Lambda g_{\mu\nu}}_{\text{negative}} = 0$$

Conformal sym	$\longleftrightarrow$	Isometry
DoF (central charge)	$\longleftrightarrow$	$1/G_{\text{Newton}}$

Gibbs vs Schwarzschild



BH sol of vacuum Einstein eq

$$R_{\mu\nu} + \frac{R}{2}g_{\mu\nu} + \Lambda g_{\mu\nu} = 0$$

negative

Fluctuation/Dissipation



Radiation/Horizon

Entropy



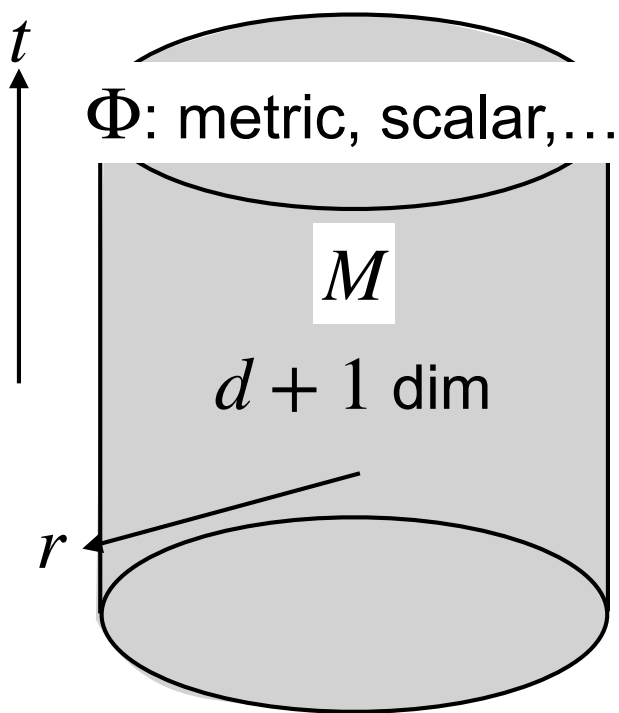
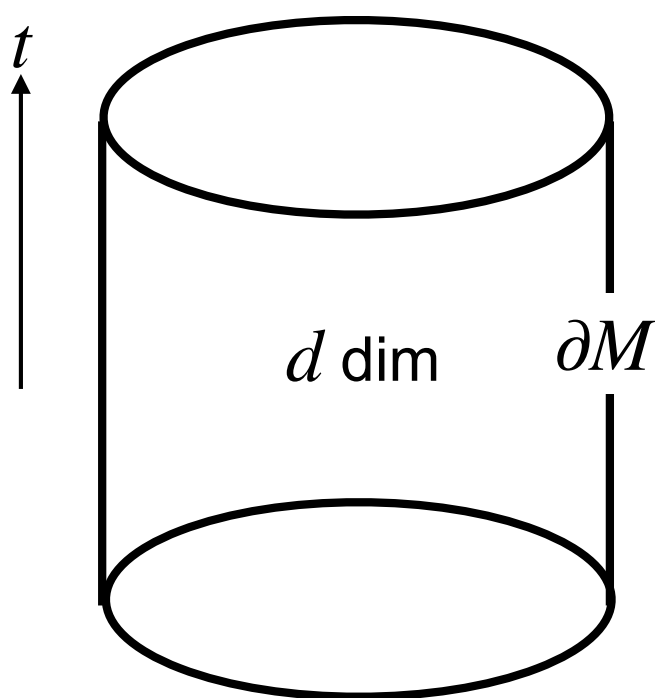
Horizon area

The GKP-W dictionary

AdS/CFT dictionary

$$\int \mathcal{D}\phi e^{iS_{\text{CFT}}[\partial M; \phi] + i \int d^d x J(x) O_{\Delta}(x)} = \int \mathcal{D}\Phi e^{iS_{\text{AdS}}[M; \Phi]} \Big|_{\Phi \sim r^{\Delta-d} J} \quad \begin{array}{l} r \rightarrow \infty \\ \downarrow \end{array}$$

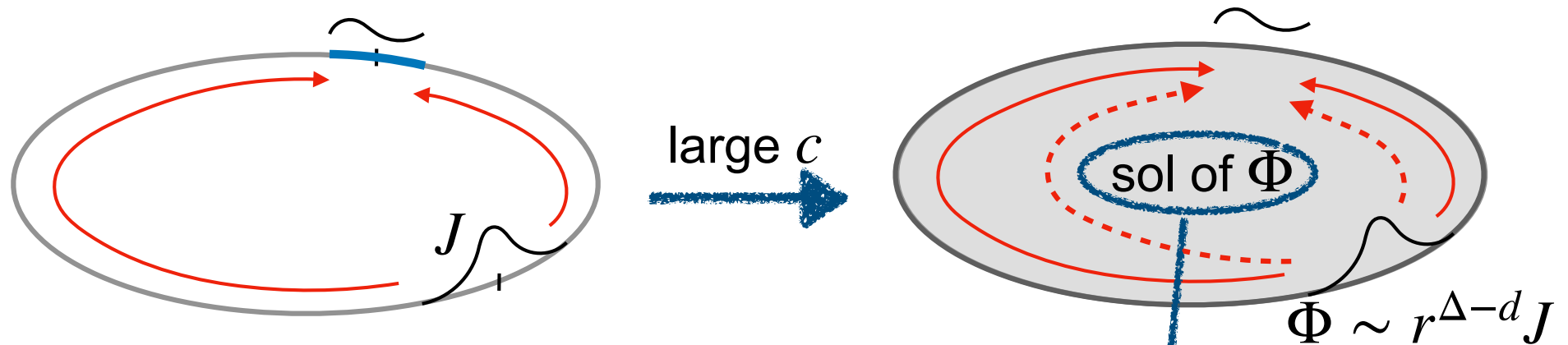
$$\stackrel{\text{large } c}{\simeq} e^{iS_{\text{AdS}}[M; \Phi_{\text{cl}}]} \Big|_{\Phi \sim r^{\Delta-d} J}$$



M is flexible : SK contour, replica manifold, ...

Exercise: get CFT 2pt function from AdS

Want to know: $\langle O \rangle_J$ (linear response)

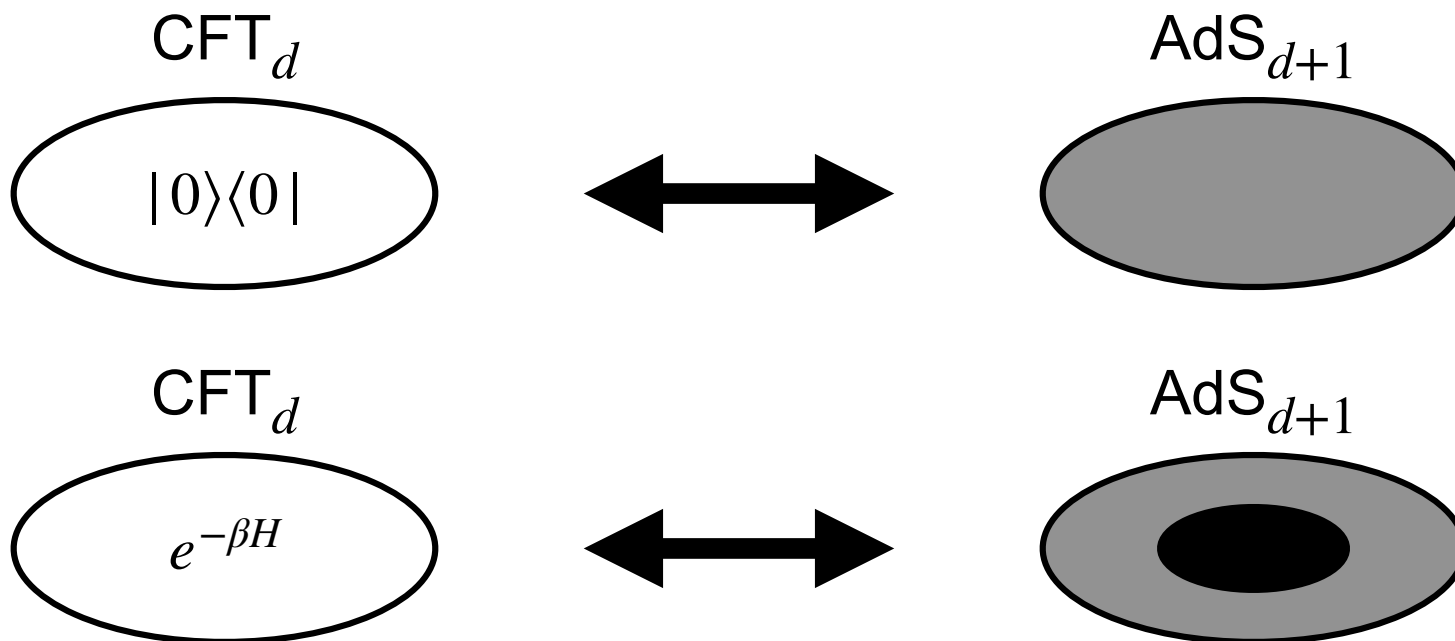


$$\int \mathcal{D}\phi e^{iS_{\text{CFT}}[\partial M; \phi] + i \int d^d x J(x) O_{\Delta}(x)} \stackrel{\text{large } c}{\simeq} e^{iS_{\text{AdS}}[M; \Phi_{\text{cl}}]} \Big|_{\Phi \sim r^{\Delta-d} J}$$

$$\frac{\delta}{\delta J}$$

$$\langle O \rangle_J = \int d^d y G_R(x - y) J(y) + \mathcal{O}(J^2)$$

Foundation: closed AdS/CFT



AdS/CFT dictionary

$$\int \mathcal{D}\phi e^{iS_{\text{CFT}}[\partial M; \phi] + i \int d^d x J(x) O_{\Delta}(x)} = \int \mathcal{D}\Phi e^{iS_{\text{AdS}}[M; \Phi]} \Big|_{\Phi \sim r^{\Delta-d} J} \quad \begin{array}{l} r \rightarrow \infty \\ \downarrow \end{array}$$

$$\stackrel{\text{large } c}{\simeq} e^{iS_{\text{AdS}}[M; \Phi_{\text{cl}}]} \Big|_{\Phi \sim r^{\Delta-d} J}$$

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A model of open CFT

$$S_{\text{tot}} = S_{\text{CFT}}[\phi] + \underbrace{\sum_n \int d^d x \left[-\frac{1}{2}(\partial\psi_n)^2 - \frac{m_n^2}{2}\psi_n^2 \right]}_{I_{\text{bath}}[\psi]} + g_n \psi_n \underbrace{O[\phi]}_{\text{CFT primary}}$$

Initial state: $|0\rangle\langle 0| \otimes \prod_n e^{-\beta H_n}$

CFT vac
bath Gibbs

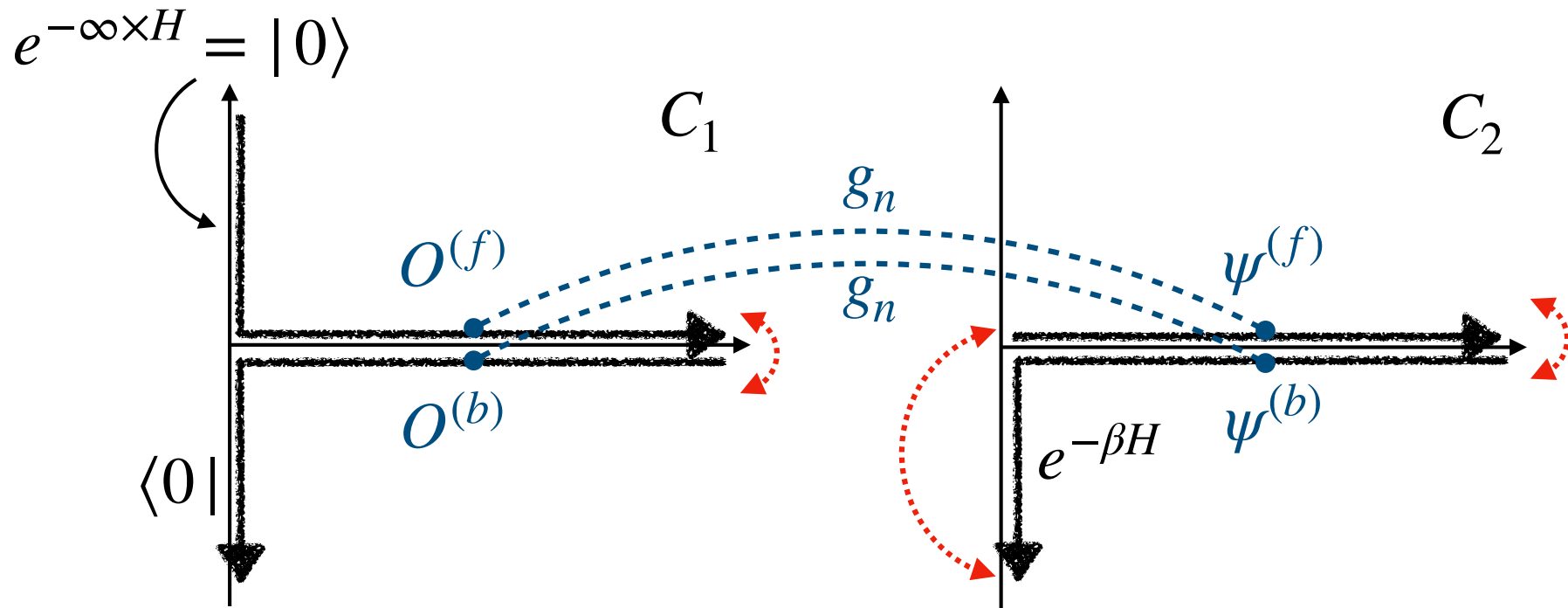
For simplicity

SK path integral

$$Z = \int \mathcal{D}\phi \mathcal{D}\psi \exp \left[iS_{\text{CFT}}[\phi; C_1] + S_{\text{bath}}[\psi; C_2] \right.$$

$$\left. + i \int_{\mathbb{R}} d^d y \sum_n g_n (\psi_n^{(f)} O^{(f)} - \psi_n^{(b)} O^{(b)}) \right]$$

$$\rho_0 = |0\rangle\langle 0| \otimes \prod_n e^{-\beta H_n}$$



Influence functional

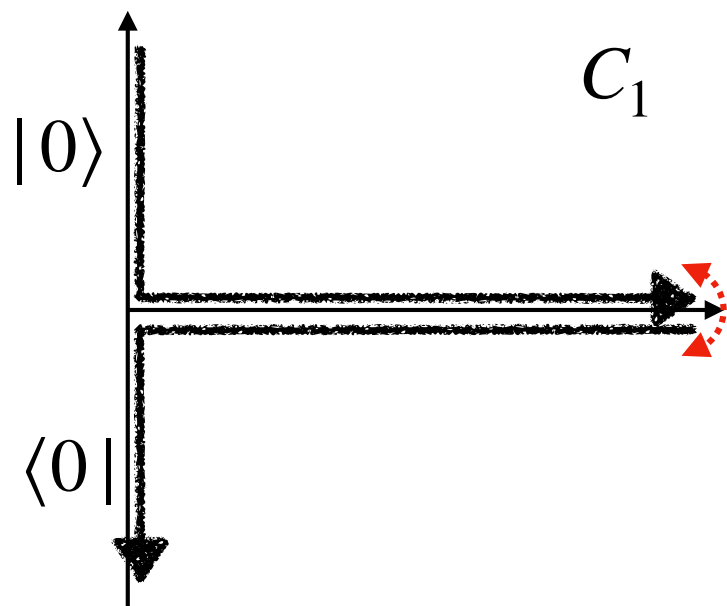
$$Z = \int \mathcal{D}\phi \, e^{iS_{\text{CFT}}[C_1; \phi]} \exp \left[- \int d^d x d^d y \left\{ g(x) g(y) \right. \right. \\ \left. \left. \times \left(2i O_a(x) \eta(x-y) \Theta(x^0 - y^0) O_r(y) + \frac{1}{2} O_a(x) \nu(x-y) O_a(y) \right) \right\} \right]$$

t-dependent coupling.
Switch on/off

$$O_r := \frac{O^{(f)} + O^{(b)}}{2} \quad O_a := O^{(f)} - O^{(b)}$$

η : bath retarded Green function (dissipation)

ν : bath Wightman function (fluctuation)



Fluctuation-Dissipation thm:

$$\tilde{\nu}(k) = -2 \coth \frac{k^0 \beta}{2} \text{Im } \tilde{\eta}(k)$$

Lagrange multipliers to apply dictionary

AdS/CFT dictionary

$$\int \mathcal{D}\phi e^{iS_{\text{CFT}}[\partial M; \phi] + i \int d^d x J(x) O_{\Delta}(x)} = \int \mathcal{D}\Phi e^{iS_{\text{AdS}}[M; \Phi]} \Big|_{\Phi \sim r^{\Delta-d} J}$$

$$O_r := \frac{O^{(f)} + O^{(b)}}{2}$$

$$O_a := O^{(f)} - O^{(b)}$$

$$Z = \int \mathcal{D}\phi \mathcal{D}\chi \mathcal{D}\xi e^{iS_{\text{CFT}}[C_1; \phi] - i \int d^d x \chi_r (\xi_r - O_r) - i \int d^d x \chi_a (\xi_a - O_a)}$$

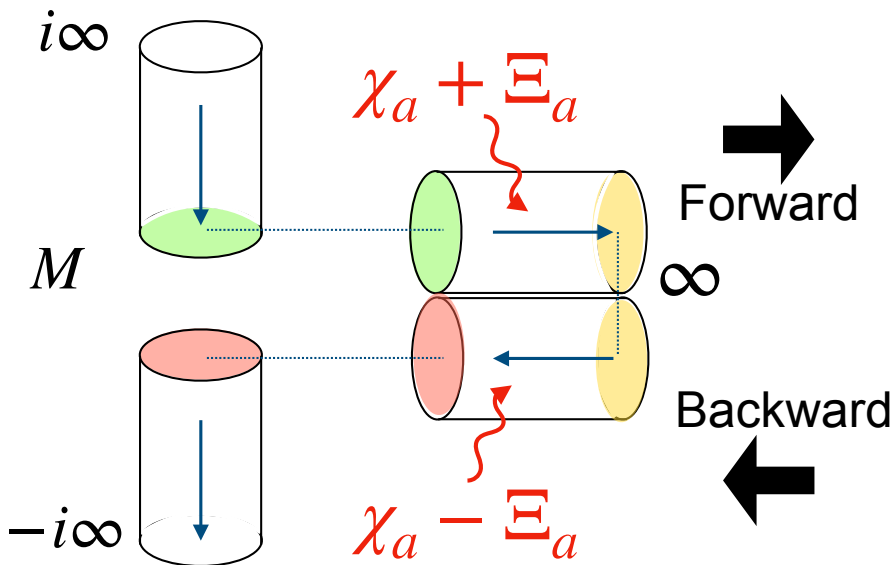
$$\times \exp \left(- \int d^d x d^d y \left[2i \xi_a(x) \mathcal{H}(x, y) \xi_r(y) + \frac{1}{2} \xi_a(x) \mathcal{N}(x, y) \xi_a(y) \right] \right)$$

$$\mathcal{H}(x, y) := g(x)g(y)\eta(x-y)\Theta(x^0 - y^0)$$

$$\mathcal{N}(x, y) := g(x)g(y)\nu(x-y)$$

The gravity dual

$$\begin{aligned}
 Z &= \int \mathcal{D}\Phi \mathcal{D}\chi \mathcal{D}\xi e^{iS_{\text{AdS}}[M, \Phi] - i \int d^d x (\chi_r \xi_r + \chi_a \xi_a)} \\
 &\quad \times \exp \left(- \int d^d x d^d y \left[2i \xi_a(x) \mathcal{H}(x, y) \xi_r(y) + \frac{1}{2} \xi_a(x) \overbrace{\mathcal{N}(x, y) \xi_a(y)}^{\mathcal{N} * \xi_a} \right] \right) \\
 &= \int \mathcal{D}\Phi \mathcal{D}\chi_a \mathcal{D}\xi_a e^{iS_{\text{AdS}}[M, \Phi] - (\xi_a, \mathcal{N} * \xi_a)/2 - (\chi_a, \mathcal{N}^{-1} * \chi_a)/2} \\
 &\quad \text{with } \Phi^{(f/b)} \sim \left[\chi_a \pm \underbrace{\left(i \mathcal{H}^T * \{ \mathcal{N}^{-1} * \chi_a \} - \mathcal{H}^T * \xi_a \right)}_{=: \Xi_a} \right] r^{\Delta-d}
 \end{aligned}$$



$$\begin{aligned}
 \mathcal{H}(x, y) &:= g(x)g(y)\eta(x-y)\Theta(x^0-y^0) \\
 \mathcal{N}(x, y) &:= g(x)g(y)\nu(x-y)
 \end{aligned}$$

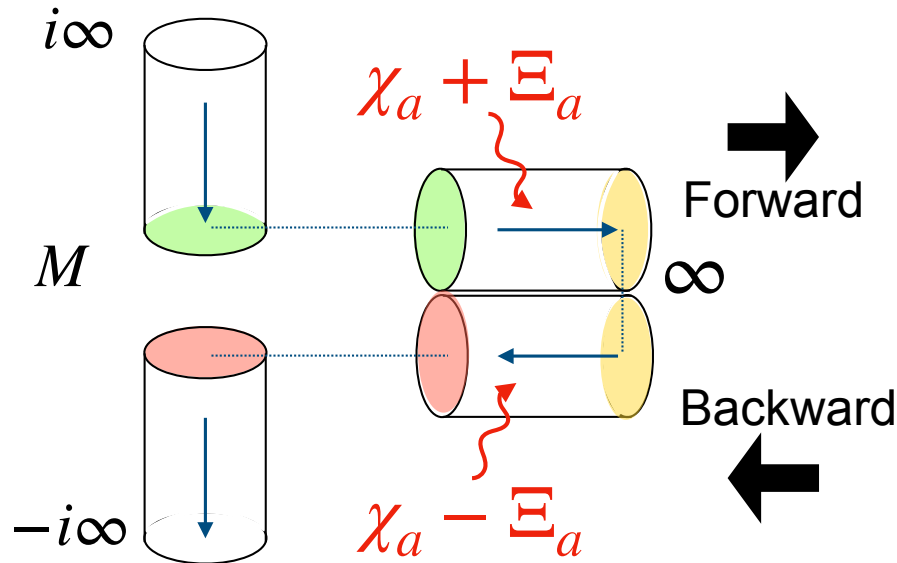
$$\tilde{\nu}(k) = -2 \coth \frac{k^0 \beta}{2} \text{Im } \tilde{\eta}(k)$$

Open AdS/CFT dictionary

$$Z = \int \mathcal{D}\phi \, e^{iS_{\text{CFT}}[C_1; \phi]} \exp \left[- \int d^d x d^d y \left\{ g(x)g(y) \right. \right. \\ \left. \left. \times \left(2iO_a(x)\eta(x-y)\Theta(x^0-y^0)O_r(y) + \frac{1}{2}O_a(x)\nu(x-y)O_a(y) \right) \right\} \right] \quad \text{CFT}$$

$$= \int \mathcal{D}\Phi \mathcal{D}\chi_a \mathcal{D}\xi_a \, e^{iS_{\text{AdS}}[M, \Phi] - (\xi_a, \mathcal{N}^* \xi_a)/2 - (\chi_a, \mathcal{N}^{-1} \chi_a)/2} \quad \text{AdS}$$

$$\Phi^{(flb)} \sim \left[\chi_a \pm \left(i\mathcal{H}^T * \{ \mathcal{N}^{-1} * \chi_a \} - \mathcal{H}^T * \xi_a \right) \right] r^{\Delta-d}$$



High-temperature bath

Integrate out

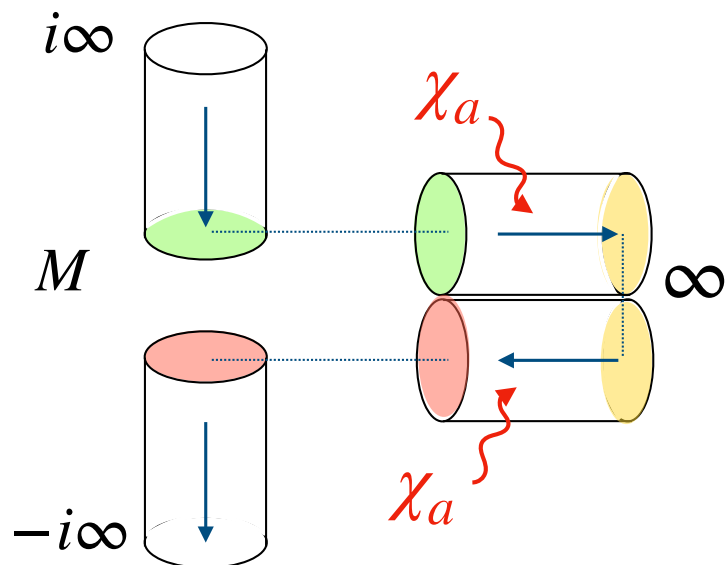
$$Z = \int \mathcal{D}\Phi \mathcal{D}\chi_a \mathcal{D}\xi_a e^{iS_{\text{AdS}}[M, \Phi] - (\xi_a, \mathcal{N}^* \xi_a)/2 - (\chi_a, \mathcal{N}^{-1} * \chi_a)/2}$$

$$\Phi^{(f/b)} \sim \left[\chi_a \pm \left(i \cancel{\mathcal{H}^T} * \{ \mathcal{N}^{-1} * \chi_a \} - \cancel{\mathcal{H}^T} * \xi_a \right) \right] r^{\Delta-d}$$

$$\mathcal{H}(x, y) := g(x)g(y)\eta(x-y)\cancel{\Theta}(x^0 - y^0)$$

$$\mathcal{N}(x, y) := g(x)g(y)\nu(x-y)$$

$$\tilde{\nu}(k) = -2 \coth \frac{k^0 \beta}{2} \text{Im } \tilde{\eta}(k) \quad \nu \gg \eta$$



$$Z = \int \mathcal{D}\Phi \mathcal{D}\chi_a e^{iS_{\text{AdS}}[M, \Phi] - (\chi_a, \mathcal{N}^{-1} * \chi_a)/2}$$

with $\Phi^{(f/b)} \sim \chi_a r^{\Delta-d}$

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JT gravity with scalar

$$S_{\text{AdS}} := S_g[g, \phi] + S_m[g, \Phi] + (\text{counter terms})$$

u : boundary time

$$(t, z) = (t(u), z(u))$$

$$S_g := -\frac{1}{16\pi G} \left[\int_M d^2x \sqrt{-g} \phi (R + 2) + 2 \int_{\partial M} du \sqrt{-h} \phi (K - 1) \right]$$

$$S_m := -\frac{1}{2} \int_M d^2x \sqrt{-g} (\partial_\mu \Phi \partial^\mu \Phi + m^2 \Phi^2)$$

$$m^2 := \Delta(\Delta - 1)$$

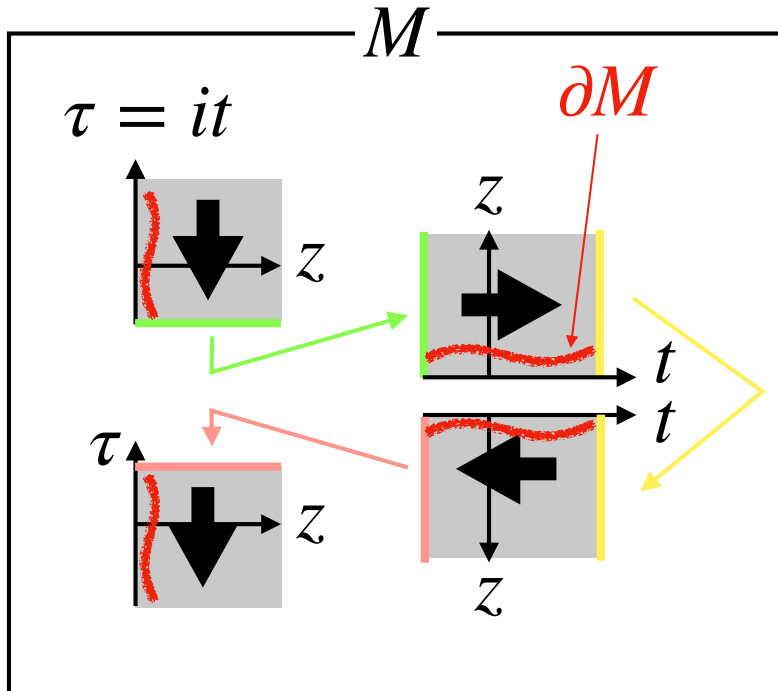
$\delta(R + 2)$ by $\mathcal{D}\phi$
 $\Rightarrow g_{\mu\nu}$: pure AdS_2

$$Z = \int \mathcal{D}g \mathcal{D}\phi \mathcal{D}\Phi \mathcal{D}\lambda e^{iS_{\text{AdS}}[M, \Phi] - (\lambda, \mathcal{N}^{-1*} \lambda)/2}$$

$$\Phi^{(f/b)}|_{\partial M} = \lambda(u) \epsilon^{1-\Delta} \quad \phi^{(f/b)}|_{\partial M} = \epsilon^{-1}$$

$$g_{\mu\nu}^{(f/b)}|_{\partial M} dx^\mu dx^\nu = -\epsilon^{-2} du^2$$

$$z(u) = \epsilon t'(u) + \mathcal{O}(\epsilon^3)$$



Integration of Φ, λ

$$Z = \int \mathcal{D}t e^{\frac{iS_{\text{Sch}}[t_f] - iS_{\text{Sch}}[t_b] + iW[t_f, t_b]}{\text{pure JT gravity}}}$$

$$S_{\text{Sch}}[t] := -\frac{1}{8\pi G} \int du \left(\frac{\ddot{t}}{\dot{t}} - \frac{3}{2} \left(\frac{\ddot{t}}{\dot{t}} \right)^2 \right)$$

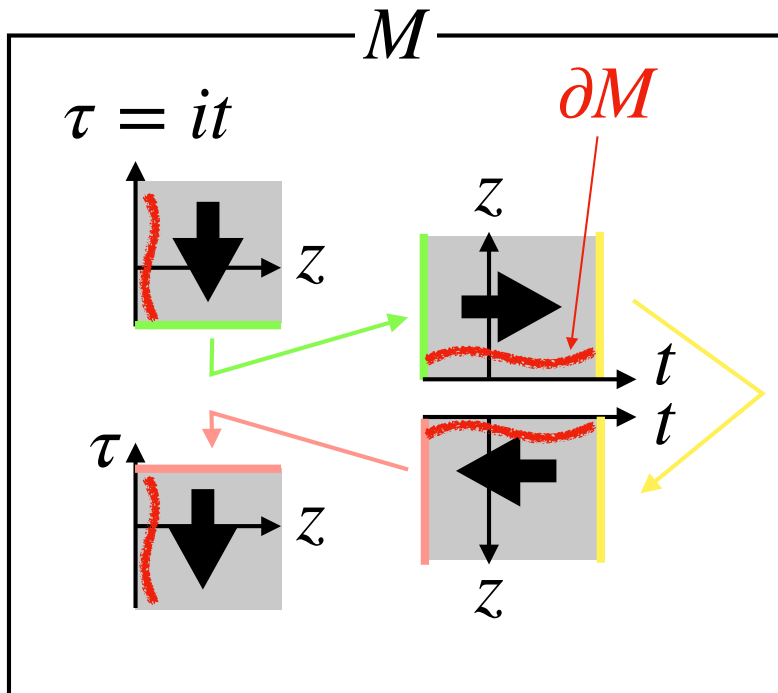
$$e^{iW[t_f, t_b]} := \int \mathcal{D}\lambda e^{iS_{m, \text{eff}}[t_f, t_b; \lambda] - (\lambda, \mathcal{N}^{-1} * \lambda)/2}$$

Solve Φ EOM on AdS_2

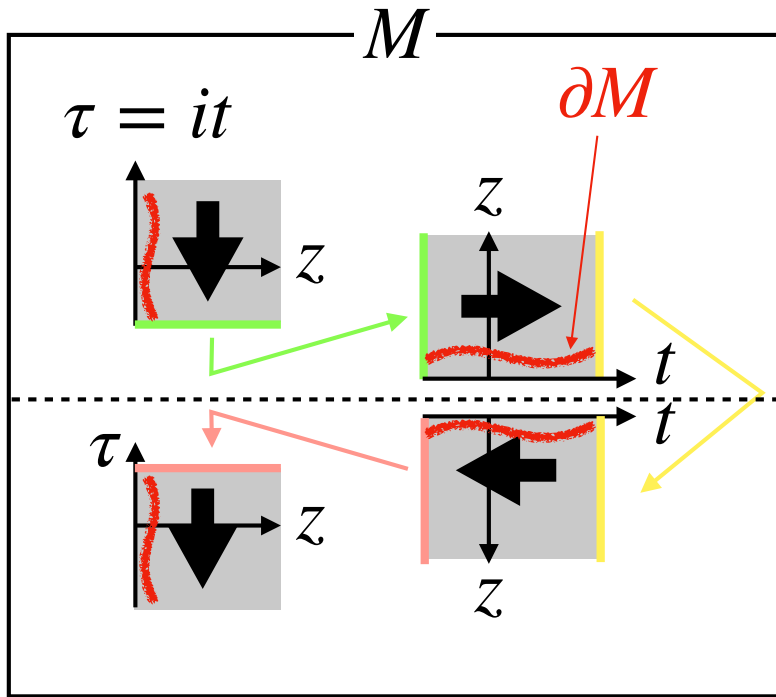
$$iS_{m, \text{eff}}[t_f, t_b; \lambda] \propto - \int_0^\infty du_1 \text{Pf} \int_0^\infty du_2 \lambda(u_1) \lambda(u_2)$$

$$\times \left[\frac{e^{-i\pi\Delta} (\dot{t}_f(u_1) \dot{t}_f(u_2))^\Delta}{|t_f(u_1) - t_f(u_2)|^{2\Delta}} + \dots \right]$$

Now, just a theory of embedding $t(u)$



EOM for embedding t



nothing excited before bath attachment

Ansatz: $\tau(u_E) = u_E$, $t_f(u) = t_b(u)$

$\mathbb{Z}_2$ symmetry

$$\mathcal{N}(x, y) := g(x)g(y)\nu(x - y)$$

$$\frac{\ddot{t}(u)}{\dot{t}(u)} - \frac{3}{2} \left(\frac{\ddot{t}(u)}{\dot{t}(u)} \right)^2 = i \text{ Pf} \int du' \left[\frac{2\Delta(\dot{t}(u)\dot{t}(u'))^\Delta}{|t(u) - t(u')|^{2\Delta+1}} \mathcal{N}(u, u') + \dots \right]$$

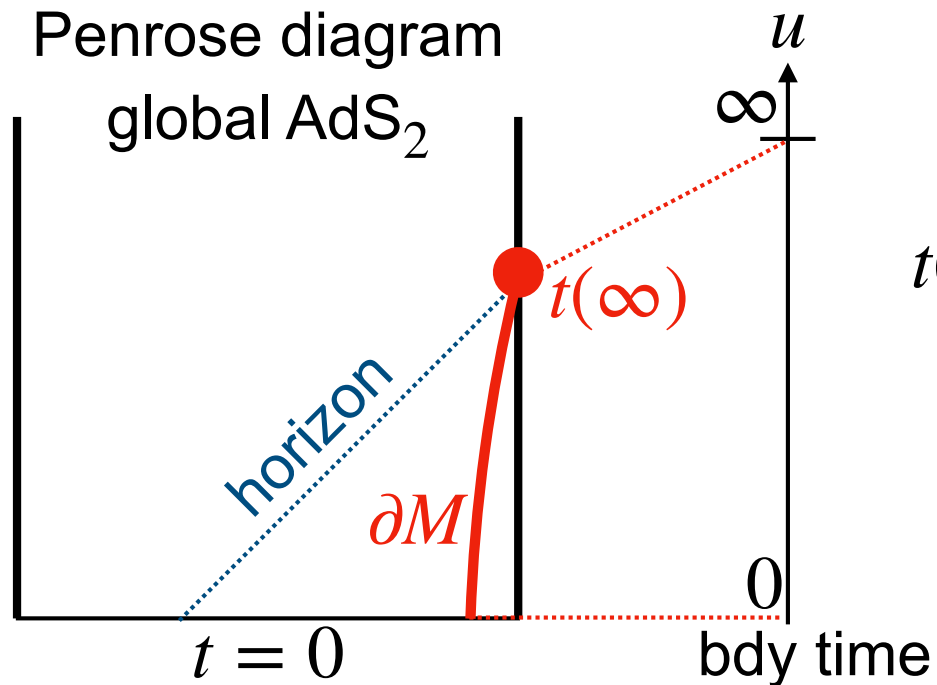
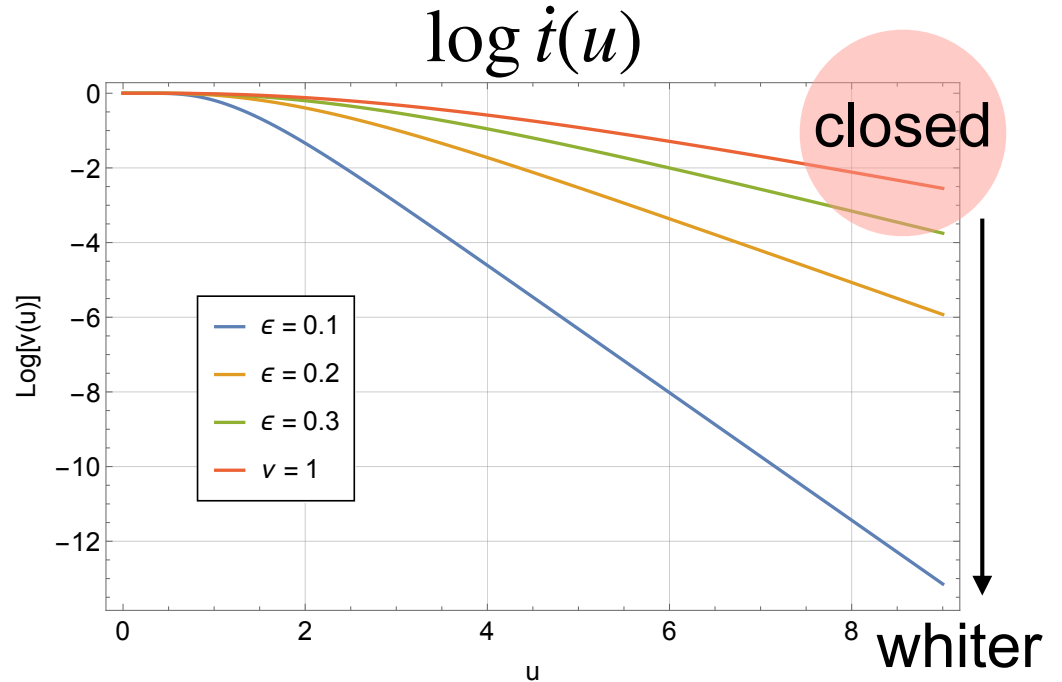
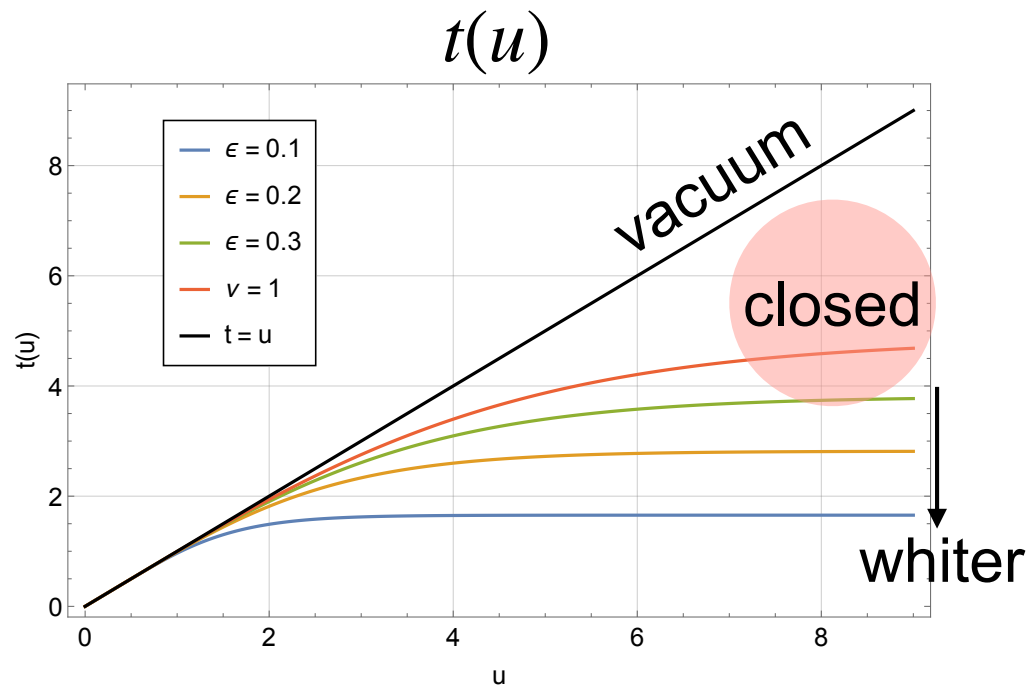
System-bath coupling

$$g(u) = \begin{cases} u^3(1-u)^3 & u \in [0,1] \\ 0 & u \notin [0,1] \end{cases}$$

Noise

$$\nu(u) = \frac{\epsilon}{\pi} \frac{1}{\epsilon^2 + u^2} \xrightarrow{\epsilon \rightarrow +0} \delta(u)$$

BH formation is confirmed!



Eternal AdS_2 BH sol

$$t(u) = \frac{\beta}{\pi} \tanh \left(\frac{\pi}{\beta} u \right)$$

$$\sim \frac{\beta}{\pi} \left(1 - 2e^{-2\pi u/\beta} \right) \quad (u \rightarrow \infty)$$

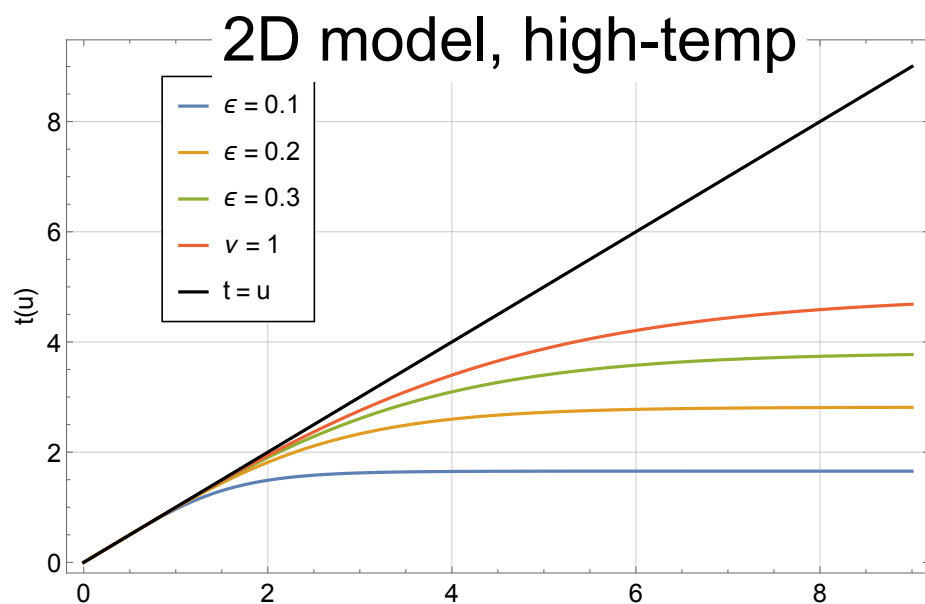
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Recipe for holographic BHs

Just consider this dynamics

$$Z = \int \mathcal{D}\Phi \mathcal{D}\chi_a \mathcal{D}\xi_a e^{iS_{\text{AdS}}[M, \Phi] - (\xi_a, \mathcal{N}^* \xi_a)/2 - (\chi_a, \mathcal{N}^{-1} \chi_a)/2}$$



Future work

- higher dimensions
- low-temperature
- comparison with Lindbladian SYK

